ADEKUNLE AJASIN UNIVERSITY

P.M.B 001, AKUNGBA-AKOKO, ONDO STATE, NIGERIA

FACULTY OF COMPUTING

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DESIGN AND DEVELOPMENT OF A SECURE EXAMINATION VERIFICATION SYSTEM USING QR CODE TECHNOLOGY

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#	NAME	ID	ROLE
1	AGWUNOBI SOMTOCHUKWU BRIGHT	220404008	Member
2	OLATUNJI JUBRIL TEMITOPE	200404169	Member
3	ADEBOWALE KOLAWOLE JOSHUA	220404170	Member
4	UBONG VICTORY PEACE	220404107	Member
5	OLUWATOMIWA OLUMOFE	170404081	Member
6	OJEKUNLE BOLUWATIFE	220404256	Member

PROJECT SUPERVISOR:

DR. OGBEIDE

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND TO THE STUDY

The integrity of academic examinations is foundational to the credibility of university education. When the processes governing who sits an examination can be circumvented, the value of every qualification that institution awards comes into question. At Adekunle Ajasin University and across Nigerian universities more broadly, the current examination admission process involves two distinct manual steps, each operating without cryptographic or real-time verification, and each creating its own category of risk (Owolabi et al., 2019; Adebayo et al., 2022).

The process begins when a student pays their school fees through the Remita electronic payment platform, which issues a Retrieval Reference (RRR) number and a printed receipt as proof of payment. Before sitting any examination, the student must travel to their faculty or department building carrying both the Remita receipt and a valid student ID card that contains their personal and academic information including course registration details. At Adekunle Ajasin University, a Computer Science student in the Faculty of Computing goes to Glasshouse, the faculty building, for this step. There, a staff member or examiner on duty reviews the documents, confirms the details by sight, and issues the student a handwritten paper confirming the amount paid. This paper is signed by the examiner on duty and printed on distinctive purple stock, making it the recognisable currency of examination clearance across the university. The student then carries this purple paper to the examination venue, where it serves as the primary admission document presented at the hall entrance.

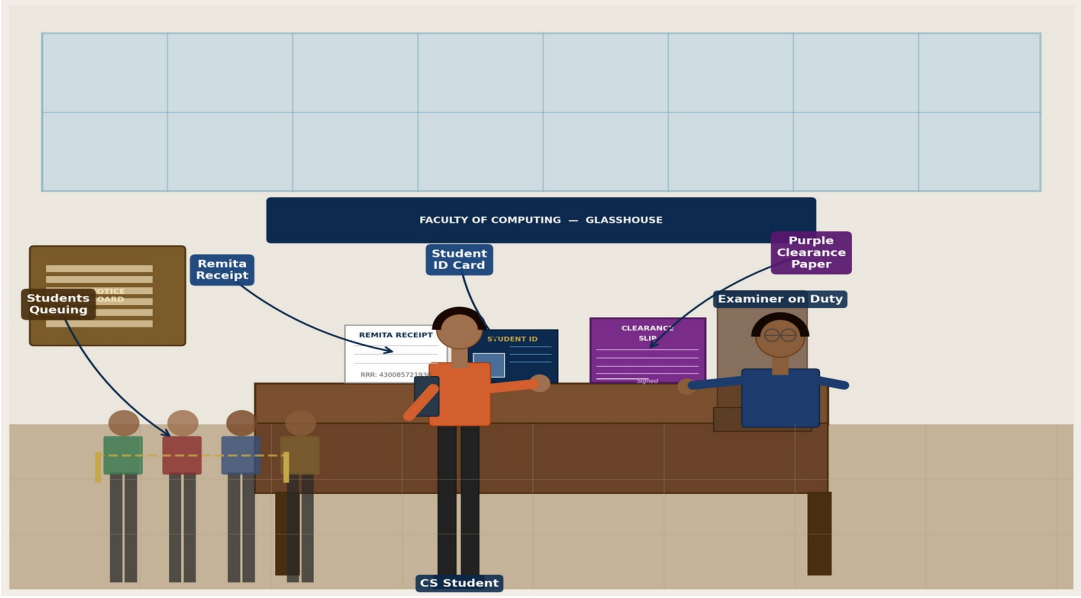

Figure 1: A student presenting a Remita receipt and student ID to the examiner on duty at Glasshouse, the Faculty of Computing building at Adekunle Ajasin University. The examiner reviews the documents and issues the purple clearance paper, which the student carries to the examination venue.

This two-step process has real, documented problems. The faculty building becomes a physical bottleneck in the days before and on the morning of examinations, with large numbers of students converging on a single building to obtain a document that could only be issued there. At Glasshouse during examination periods, the resulting overcrowding disrupts the building's normal functioning and places significant strain on the small number of staff members handling the queue. Students who arrive late or encounter any complication at the faculty building risk missing the start of their examination entirely (Ezeudu & Chiaha, 2018).

The administrative inefficiency of the process is compounded by the fact that the purple paper itself solves none of the fundamental verification problems. It is a handwritten document with no photograph, no biometric element, and no cryptographic signature. Any student can present someone else's purple paper at the examination gate, and the invigilator has no reliable mechanism for detecting the substitution beyond a visual comparison of faces. The paper

confirms that someone paid the correct amount; it does not confirm that the person holding it is the person who paid (Petters & Okon, 2014). Impersonation remains technically straightforward.

QR code technology, combined with real-time payment verification through Remita's API, offers a way to collapse both steps into a single cryptographically secured credential issued at the point of online registration. Instead of a student making a separate trip to the faculty building, the system verifies the Remita payment programmatically, binds the confirmed payment record to the student's identity and passport photograph, encrypts the combined payload, and encodes it as a QR code for printing. At the examination gate, an invigilator scans the printed code, and the server decrypts it, validates its signature, displays the student's photograph for visual confirmation, and marks the token as permanently consumed in real time. No faculty building visit. No handwritten paper. No window for impersonation through a document that carries no biometric link to its holder (National Institute of Standards and Technology, 2023; Remita Payment Services, 2023).

1.2 Statement of the Problem

The current examination clearance process at Nigerian universities presents a cluster of interrelated failures rooted in its two-step, document-based architecture. Together they constitute a structural breakdown that issuing a better-designed paper form cannot resolve.

First, the faculty building step creates physical overcrowding and disrupts normal operations. When large cohorts of students must physically travel to a single building to obtain the purple clearance paper, the resulting concentration of foot traffic overwhelms the available space and the staff managing the queue. At Glasshouse, the Faculty of Computing building, this congestion is a predictable and recurring problem before every examination period. Students who cannot reach the front of the queue in time risk missing the examination entirely, a consequence that falls hardest on those who travel from off-campus accommodation (Ezeudu & Chiaha, 2018).

Second, the process is administratively inefficient. The extra trip to the faculty building adds a mandatory stage between payment and examination that serves no verification function that could not be performed more accurately and quickly by a system with programmatic access to

Remita's records. Staff at the faculty building are occupied producing handwritten documents for hours before every examination when that time could be allocated to academic work. As student enrolment grows, this inefficiency scales proportionally (Federal Ministry of Education, 2022).

Third, the purple paper does not solve the impersonation problem it was intended to address. The paper is handwritten and carries no photograph, no biometric element, and no cryptographic signature linking it to a specific individual. Any student who obtains or copies another student's purple paper can present it at the examination gate, and the invigilator has no reliable mechanism for detecting the substitution. Visual comparison of faces against a name on a handwritten slip is not a dependable security control in a large institution where invigilators do not personally know every student (Petters & Okon, 2014).

Fourth, the faculty building verification step is itself vulnerable to document fraud. The Remita receipt presented at Glasshouse can be photocopied and altered before the student arrives, with names, amounts, or RRR numbers modified using basic image editing software (Okoye & Ezeanyanso, 2013). The examiner on duty has no access to Remita's database and cannot confirm whether the receipt is genuine. The purple paper they issue on the basis of that receipt therefore inherits the fraud risk of the document it was issued against.

Fifth, cognitive fatigue compounds the verification failures at both stages. Examiners processing large queues of students at the faculty building, and invigilators admitting students at the examination gate, both experience measurably reduced attention to anomalies after extended periods of repetitive document checking (Kahneman, 2011). This is not a personnel shortcoming; it is a predictable consequence of designing security-critical tasks around unsupported human attention.

Sixth, neither stage produces a record suitable for retrospective investigation. The handwritten queue at the faculty building and the signed attendance sheet at the examination gate are both vulnerable to incompleteness, illegibility, and tampering. When impersonation allegations arise after an examination, the investigation is hampered from the outset by the absence of any

timestamped, cryptographically verifiable record of what documents were presented, by whom, and when (Federal Ministry of Education, 2022).

This project addresses all six dimensions. The faculty building bottleneck is eliminated by moving clearance to an online registration workflow completed before examination day. Document fraud is closed by making programmatic Remita API verification a prerequisite for credential issuance. Impersonation is countered by binding the credential to a server-retrieved photograph displayed at the gate. Cognitive fatigue is removed as a variable by replacing human document judgement with cryptographic checking. Every admission decision leaves an immutable, timestamped audit record.

1.3 Aim and Objectives of the Study

Aim of the study

The aim of this study is to design and develop a secure examination admission verification system that uses cryptographically protected, one-time-use QR code tokens to replace the current two-step manual clearance process (faculty building visit followed by handwritten purple paper) with a single digitally verifiable credential, with specific implementation targeting the examination context of Adekunle Ajasin University, Akungba-Akoko.

Objectives

The specific objectives of this study are to:

i. Design and implement a secure student registration portal that validates matriculation numbers, captures passport photographs, and stores verified student records in a relational database.
ii. Integrate the system with the Remita payment verification API to confirm, in real time, that a submitted RRR number corresponds to a successful examination fee payment, ensuring that QR credential issuance is conditional on verified payment.

iii. Generate encrypted, digitally signed QR code credentials using AES-256-GCM encryption and HMAC-SHA256 digital signatures, and implement one-time-use enforcement through atomic database transactions that prevent any token from being approved more than once.
iv. Develop an examiner verification dashboard accessible through a standard web browser, using WebRTC camera integration to scan printed QR codes and display retrieved student photographs and payment confirmation records within a target latency of ten seconds.
v. Implement a comprehensive append-only audit logging system that records every verification attempt, examiner identity, timestamp, device fingerprint, and decision outcome, supporting post-examination investigation and reporting.
vi. Evaluate the system against defined success metrics covering verification speed, security robustness against forgery and reuse attacks, user interface usability, and system load performance under simulated peak examination conditions.

1.4 Methodology

The design and development methodology of the proposed system is built around three guiding principles: security by design, operational simplicity, and institutional fit. Security by design means that cryptographic guarantees, not procedural controls, form the primary defence against fraud. Operational simplicity means the system requires no specialised hardware beyond the smartphones invigilators already carry and the printers already available to students at any campus business centre, and eliminates the mandatory faculty building visit that is the current source of pre-examination congestion. Institutional fit means the system integrates with Remita, the payment gateway universities already use, and deploys on standard web hosting infrastructure. The system is structured into five principal components: the student registration module, the payment verification service, the QR generation and cryptographic layer, the examiner verification dashboard, and the administrative audit interface. The complete architecture of the proposed system is illustrated in Figure 2, showing all component relationships, data flow directions, and the boundary between internal services and external APIs.

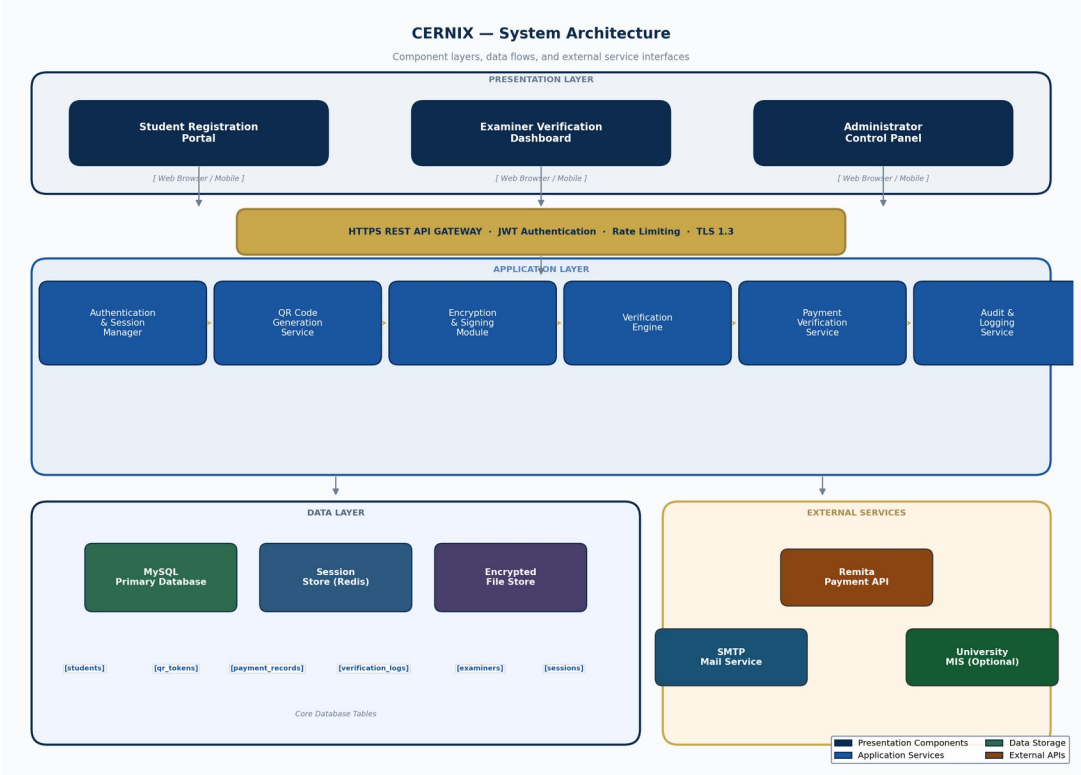

Figure 2: System architecture showing the three-layer structure, inter-service communication paths, external API boundaries, and the core database schema relationships.

1.4.1 Student Registration Module

The student registration module is the entry point for the verification workflow. A student accesses the portal and authenticates using their matriculation number, which the system validates against a pre-populated student record table sourced from the university's academic registry. The system checks for an existing active registration for the target examination session and, if one is found, redirects the student to their existing QR token rather than creating a duplicate. New registrations proceed through a structured form that captures an uploaded passport photograph, which is stored in an encrypted file store and linked to the student record by a hashed reference. This photograph is later retrieved and displayed on the examiner's dashboard during gate verification, providing the visual identity confirmation that replaces the

examiner's inspection of an ID card. The registration module produces no QR output until Remita payment verification has been completed successfully.

1.4.2 Payment Verification Integration

Payment verification is the mechanism that separates this system from a simple digital ID card. The student submits their Remita Retrieval Reference number either by manual entry or by uploading a photograph of their payment receipt, from which the system extracts the RRR using optical character recognition. The system then sends a structured HTTP request to the Remita payment verification endpoint, receiving a response that includes payment status, transaction amount, payer name, and payment date. This response is evaluated against three conditions: the payment must have been successful, the amount must correspond to the current examination registration fee, and the RRR must not already have been consumed by a previous student registration. If any condition fails, the student receives a specific error message and the registration is blocked. If all conditions are met, the payment record is stored and the RRR is marked as consumed, preventing any other student from using the same reference. Table 1 summarises the key Remita API response codes and their handling within the system.

Response Code	Status	System Handling	Student Notification
00	Payment Successful	Proceed to QR generation	Registration confirmed
01	Pending / Not Completed	Block registration	Payment not yet confirmed
02	RRR Not Found	Block; flag for review	Invalid RRR - check receipt
03	Amount Mismatch	Block registration	Incorrect fee amount paid
04	Already Consumed	Block; alert administrator	RRR already used by another student
05	Transaction Reversed	Block registration	Payment has been reversed
99	API Timeout	Queue for retry	Verification pending - retry

			in progress

Table 1: Remita API response codes and corresponding system handling logic during payment verification.

1.4.3 QR Code Generation and Cryptographic Layer

Upon successful payment verification, the system constructs a structured JSON payload containing the student's matriculation number, full name, examination session identifier, a SHA-256 hash of their stored passport photograph, a UUID v4 token identifier unique to this registration, and a Unix timestamp recording the moment of generation. This payload is encrypted using AES-256-GCM, a symmetric authenticated encryption algorithm that simultaneously provides confidentiality and integrity: no one can read the payload without the server's private key, and any tampering with the ciphertext causes decryption to fail. The encrypted ciphertext, together with the GCM authentication tag and initialisation vector, is then signed using HMAC-SHA256, providing cryptographic proof that the QR code was generated by the legitimate university server. The resulting string is encoded into a QR code image displayed to the student with a clear print instruction. The token is simultaneously recorded in the database with a status of UNUSED, linked to the student record, the examination session, and the payment record. Figure 3 illustrates the complete QR code generation workflow from initial student login through payment verification and cryptographic processing to issuance of the printed credential.

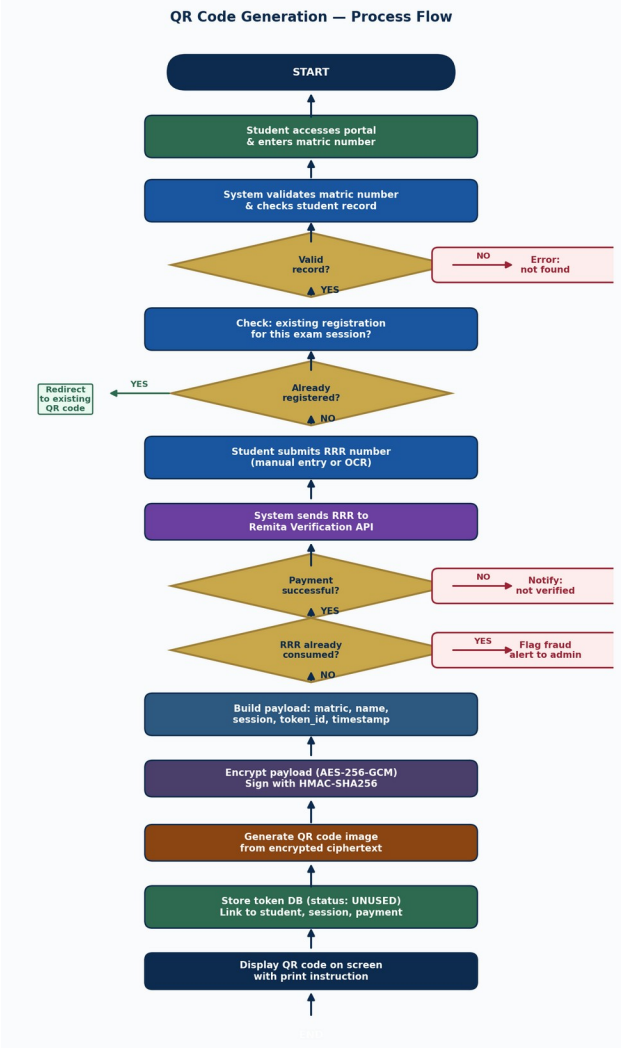

Figure 3: QR code generation process flow, showing student registration, Remita payment verification, cryptographic payload construction, and database token recording.

1.4.4 Examination-Day Verification Workflow

On the day of the examination, the student presents their printed QR code at the hall entrance. The invigilator, authenticated on the verification dashboard using their staff credentials, activates the WebRTC camera interface within their browser. Pointing the camera at the printed code causes an embedded JavaScript QR decoder to extract the ciphertext, which is immediately transmitted via HTTPS POST to the server-side verification endpoint. The server decrypts the payload using its AES-256-GCM key, validates the HMAC signature to confirm authentic origin,

and queries the database to retrieve the token's current status. If the token has not been used, the server returns the student's photograph, full name, matriculation number, department, and payment confirmation to the examiner's dashboard. The invigilator performs a visual identity check by comparing the retrieved photograph with the student standing in front of them, then either approves or rejects admission. An approval triggers an atomic database transaction that marks the token as USED and writes a timestamped log entry recording the examiner's identity, device fingerprint, and decision. This transaction completes before the approval confirmation is returned to the client, closing any window through which a simultaneous scan at another entrance could succeed. Figure 4 shows the full examination-day verification flow.

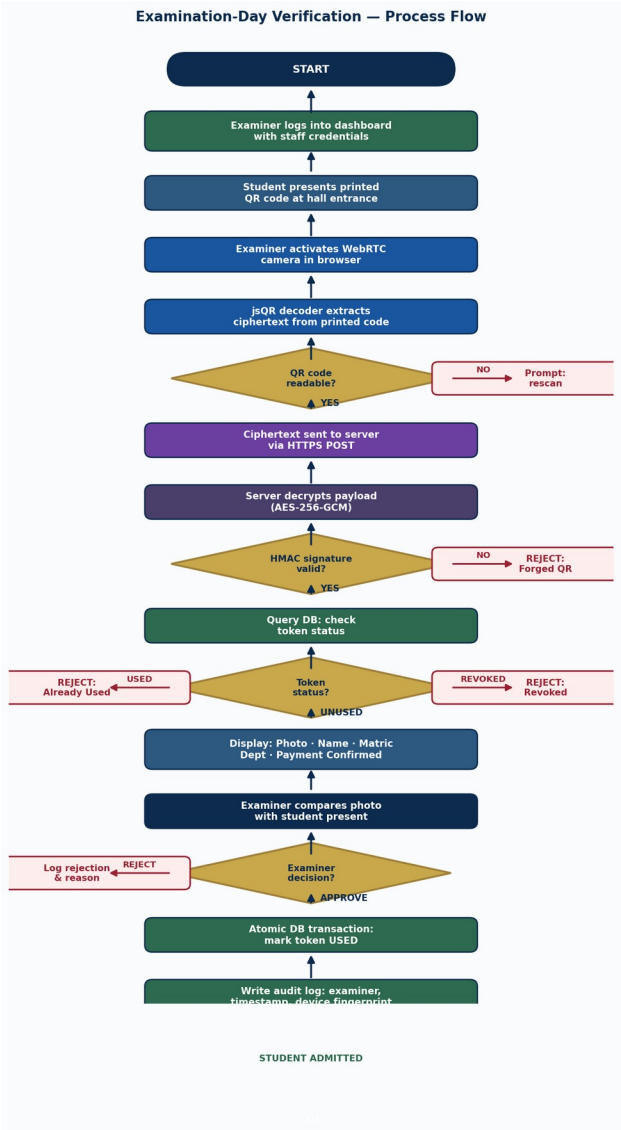

Figure 4: Examination-day verification process flow, showing QR scanning via WebRTC, server-side decryption and signature validation, database token status checking, examiner decision capture, and atomic token invalidation.

1.4.5 Examiner Authentication and Dashboard

The examiner dashboard is a role-protected web interface accessible only to accounts with examiner, supervisor, or administrator privileges. Authentication uses password-based login with bcrypt-hashed credentials, and sessions are managed through JSON Web Tokens with short expiry intervals to limit the window of credential misuse. Each examiner account is associated with a designated examination venue, and the dashboard scopes its token validation queries to that venue's active sessions. The scanning interface is designed for minimal interaction: the examiner sees a live camera feed occupying most of the screen, with a bordered target region to guide QR code placement. Upon successful decoding, the student photograph and details appear in an overlay with prominent Approve and Reject buttons. The entire visual interaction is completable in under ten seconds, including the approval confirmation. The dashboard also presents a real-time session summary showing counts of approved, rejected, and duplicate-attempt cases, giving supervisors situational awareness during the admission period.

1.4.6 Database Design

The database comprises seven interrelated entities that capture the complete state of the verification system. The Student entity stores core identity information, with the matriculation number as the primary key, along with full name, department reference, a link to the stored passport photograph, and contact details. The Department entity normalises department and faculty information to avoid redundancy across student records. The Examination Session entity captures the parameters of each examination event, including course code, date, time, venue, and academic session, providing the scope within which registrations and tokens are valid. The Payment Record entity stores the full Remita verification response against each submitted RRR, establishing an auditable record of the payment confirmation that underpinned each QR issuance.

The QR Token entity is the central transactional record, linking student, session, and payment, storing the encrypted payload and digital signature, and recording the token's current lifecycle status. The Verification Log entity captures every scan attempt with examiner identity, timestamp, device fingerprint, and outcome. The Examiner entity manages staff credentials, role assignments, and venue associations. Figure 5 presents the complete entity-relationship diagram for the database, showing all entities with their primary and foreign keys, attribute definitions, and cardinality relationships.

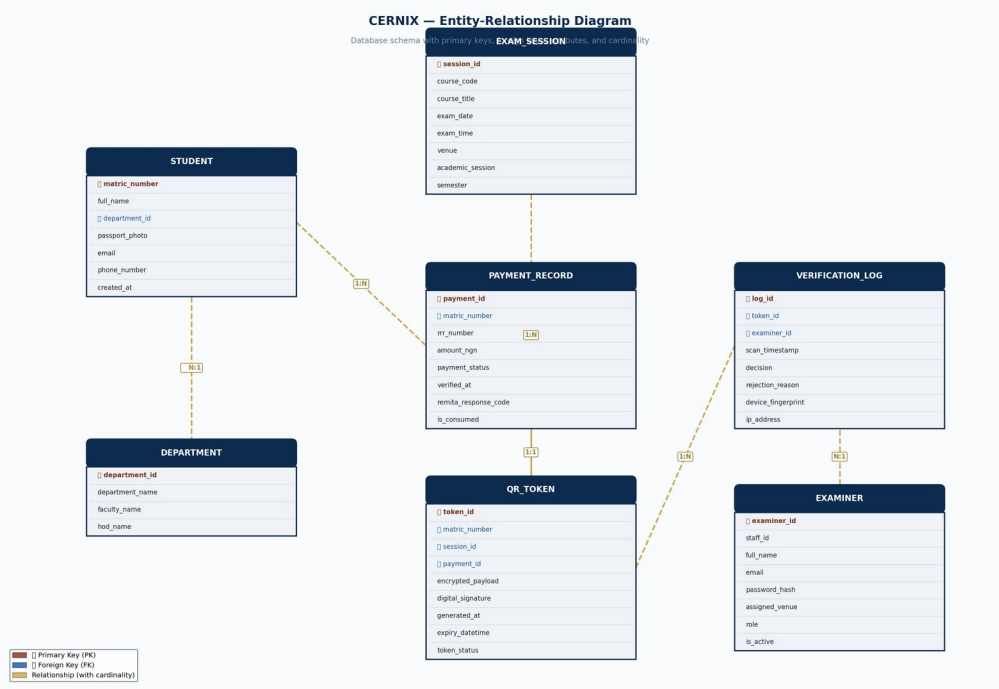

Figure 5: Entity-relationship diagram showing all seven database entities with primary keys, foreign keys, attributes, and cardinality annotations.

1.4.7 Security Architecture

Security in the proposed system is layered across the full stack, with each layer addressing a distinct attack category. The principle guiding this architecture is that no single control should be assumed infallible; each layer provides residual protection if another is somehow compromised.

Table 2 summarises the security controls deployed at each layer, the threat they address, and the implementation mechanism.

Security Layer	Threat Addressed	Control Mechanism
Transport	Interception and eavesdropping	TLS 1.3 on all API and dashboard connections
Authentication	Unauthorised examiner access	bcrypt-hashed passwords with short-lived JWTs
QR Payload	Fake QR code generation	AES-256-GCM encryption with HMAC-SHA256 signing
Token State	Photocopied or reused codes	Atomic database transaction; token marked USED at approval
Payment	Fake or recycled receipts	Live Remita API query; RRR consumption flag at registration
Identity	Impersonation	Server-retrieved photograph displayed on examiner device
Audit	Record tampering and disputes	Append-only log with timestamp, examiner identity, and device hash
Input Validation	SQL injection and XSS attacks	Parameterised queries; output escaping; Content Security Policy headers
Rate Limiting	Brute-force API attacks	Per-IP rate limits at the API gateway, 100 requests per minute

Table 2: Security controls deployed across the system stack, showing threat categories and implementation mechanisms at each layer.

1.4.8 Administrative Audit Interface

The administrative interface provides registry and examination officers with a structured view of every examination session's verification history. In addition to real-time session summaries showing total scans, approvals, rejections, and duplicate attempt counts, the admin dashboard

supports case-level investigation of flagged incidents. When a verification log entry is flagged, either automatically because the same token was scanned more than once, or manually by an examiner who suspects an irregularity, the administrator can review the full verification history for that token: every scan attempt, every outcome, every examiner who interacted with it, and every device fingerprint involved. If a case is determined to be a legitimate error, for instance a student whose code was scanned but the approving action was missed, an administrator can issue a time-limited override token signed with a separate administrative key. All administrative actions are appended to the audit log as immutable entries, preserving a complete chain of custody for every decision. Figure 6 illustrates the administrative audit and override workflow.

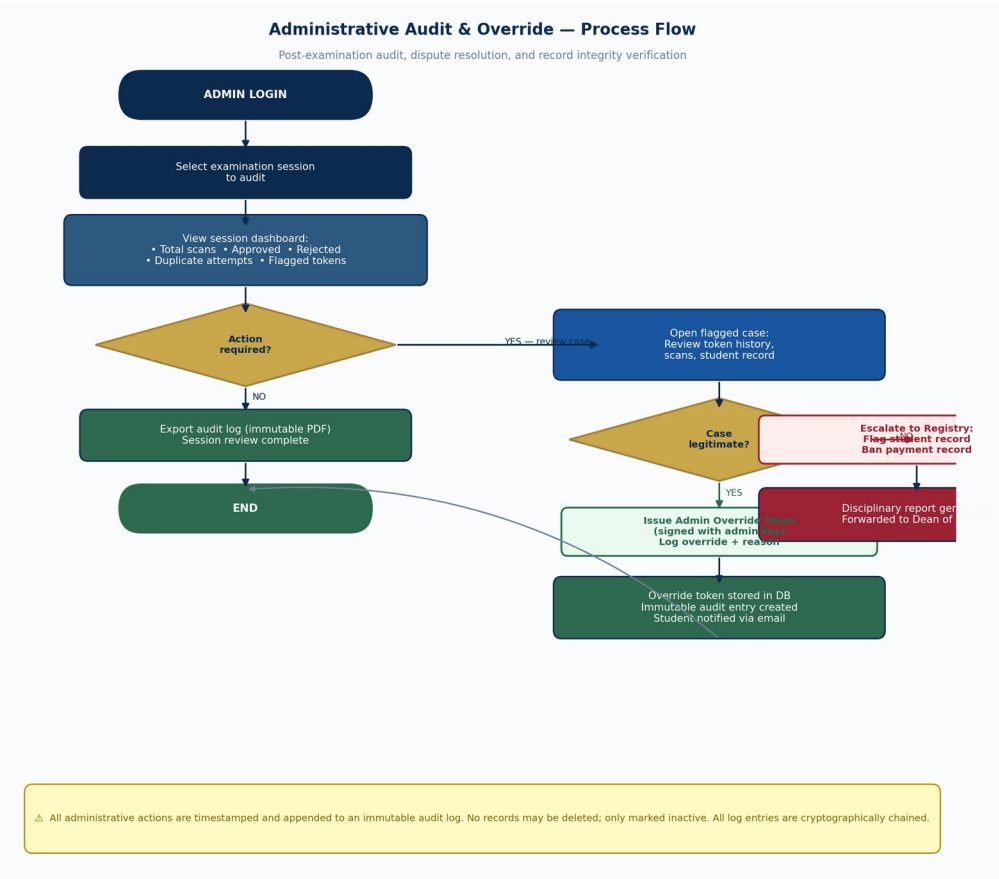

Figure 6: Administrative audit and override process flow, showing session review, flagged case investigation, legitimate override issuance, and fraud escalation pathways.

1.4.9 Development Stack and Tools

The development stack has been selected to prioritise institutional compatibility, developer familiarity, and long-term maintainability. The frontend is implemented using standard HTML5, CSS3, and vanilla JavaScript with the Tailwind CSS utility framework for responsive layout. The backend uses PHP with the Laravel framework, selected for its widespread adoption in Nigerian university hosting environments and its robust ecosystem for authentication, database abstraction, and scheduled tasks. MySQL serves as the primary relational database. Redis provides session storage for active tokens. QR code generation uses the phpqrcode library, which produces high-density QR images suitable for printing and reliable scanning from paper. Encryption is implemented using PHP's OpenSSL extension with AES-256-GCM. WebRTC camera access on the examiner's device is managed through the jsQR JavaScript library, which decodes QR patterns from video frames without requiring any app installation. Table 3 provides a complete inventory of the development tools and their roles within the system.

Component	Technology	Version Target	Purpose
Frontend	HTML5, Tailwind CSS, JavaScript	CSS v3.4	Student portal, examiner dashboard, admin interface
Backend	PHP with Laravel	Laravel 11	API routing, authentication, business logic
Database	MySQL	8.0+	Relational store for all system entities
Session Store	Redis	7.0+	JWT token management and session cache
QR Generation	phpqrcode	1.1.4	High-density QR image generation
QR Decoding	jsQR (JavaScript)	1.4.0	Browser-side QR decoding from WebRTC stream

Encryption	OpenSSL (PHP extension)	AES-256-GCM	Payload encryption and HMAC signing
Payment API	Remita HTTP REST API	v3	Real-time payment verification
Testing	PHPUnit and Pest	Latest stable	Unit, integration, and security testing
Deployment	Apache or Nginx, Ubuntu 22.04	LTS	Compatible with university hosting environments

Table 3: Development technology stack showing the tool, target version, and functional role of each component.

1.4.10 Testing Framework

The testing framework operates at three levels. Unit tests verify the correctness of individual functions: QR payload construction, encryption and decryption round-trips, HMAC signing and validation, and Remita API response parsing. Integration tests verify the full registration and verification workflows end-to-end, including database state transitions, token lifecycle progression, and concurrent scan collision handling. Security tests specifically target the forgery and reuse attack surfaces: attempting to scan a decoded and re-encoded QR code should fail signature validation; attempting to approve a token at two simultaneous sessions should be caught by the atomic transaction and produce at most one successful approval; and attempting to create a token without a verified Remita payment should be blocked at the verification layer without exception. Load tests simulate peak examination admission traffic to confirm that the verification endpoint maintains sub-ten-second response times under two hundred concurrent scan requests.

1.5 SCOPE OF THE STUDY

The system addresses the examination admission phase exclusively. It does not cover examination scheduling, question paper distribution, in-hall supervision, result processing, or grading. Payment integration covers Remita only, as it is the mandated gateway for government institutions in Nigeria. Student records are consumed from an existing registry export rather than created by the system. The project implements photograph display for visual identity confirmation and does not extend to automated facial recognition, which would introduce computational and data protection complexity beyond the project scope.

The system is a web application, ensuring compatibility across devices without requiring app installation. Internet connectivity is required at the examiner's device during verification; QR code generation occurs during the pre-examination registration window when students have connectivity available. The prototype targets a single-server deployment and does not include high-availability clustering or disaster recovery. If live Remita API credentials are unavailable during development, a mock server replicating the same request-response patterns will be used, with the expectation that live credentials would be provided for any actual institutional deployment.

1.6 Conclusion

This project addresses a structural problem affecting thousands of students and staff every semester. The current two-step clearance process, in which students travel to their faculty building to exchange a Remita receipt and ID card for a handwritten purple paper, then carry that paper to the examination venue, is slow, operationally disruptive, and incapable of preventing impersonation. The purple paper confirms that someone paid; it does not confirm that the person presenting it is that someone.

The proposed system uses technology already available to most people, integrates with the Remita payment infrastructure already in use, and eliminates the mandatory faculty building visit that is the primary source of pre-examination congestion. QR codes with cryptographic payloads replace the purple paper with a credential that is faster to issue, faster to verify, and far harder to

misuse. No visit to Glasshouse. No handwritten queue. No window for impersonation through a document that carries no biometric link to its holder.

Students reach the examination hall faster with less stress. Examiners have tools that actually verify what they are presented. The university gains a reliable, timestamped audit record for every admission decision. The theoretical and empirical foundation for this approach is established in Chapter Two, which follows.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the technologies, systems, and theoretical frameworks underpinning the design of the proposed examination admission and verification system. It examines the weaknesses of the current two-step manual clearance process used at Nigerian universities, surveys how QR code and payment verification technologies have been applied in comparable contexts globally and within Nigeria, discusses the security and access control principles the system is built on, and identifies the specific gaps in existing practice that this project addresses. The review draws on academic literature, documented deployments, and established cryptographic theory to justify the design decisions set out in Chapter Three.

2.2 Background Concepts

2.2.1 Student Identification and Verification

At Nigerian universities, examination admission currently operates as a two-stage manual process. In the first stage, a student who has paid their school fees via Remita travels to their faculty or department building carrying both the printed Remita receipt and a valid student ID card containing their personal and academic details. At Adekunle Ajasin University, a Computer Science student goes to Glasshouse, the Faculty of Computing building, for this step. An examiner on duty reviews the documents, confirms the payment details by sight, and issues a handwritten paper confirming the amount paid. This paper is signed by the examiner and printed on distinctive purple stock, which makes it the recognisable token of examination clearance across the institution. In the second stage, the student carries this purple paper to the examination venue, where an invigilator checks it at the hall entrance before admitting the student. The student ID card has served its institutional function since the first stage and is not the primary document checked at the gate; the purple paper is.

The ID card itself carries well-documented limitations as a verification instrument. Petters and Okon (2014) found that borrowed cards are a significant impersonation pathway in large Nigerian institutions where invigilators cannot personally recognise every student. Card photographs degrade over time, students' appearances change across years of study, and the physical card carries no live database connection that could confirm its validity in the moment it is presented. These weaknesses are not solved by the purple paper, which is handwritten and carries no photograph, no biometric element, and no cryptographic signature. Either document can be borrowed or copied; neither binds the credential to the specific person holding it.

2.2.2 Payment Validation in Nigerian Universities

The Remita electronic payment platform is mandated by the Federal Government for all federal institutions and widely adopted by state universities, making it the standard infrastructure for university fee collection in Nigeria (Ibrahim & Ibrahim, 2015). Remita assigns a unique Retrieval Reference (RRR) number to each transaction, and students complete payments through banks, online platforms, or mobile services. The receipt issued by Remita serves as the student's proof of payment.

At the faculty building stage of the current clearance process, the examiner on duty inspects this receipt visually. There is no system query, no API call, and no access to Remita's database. The examiner is reading a printed document and making a judgement about its authenticity based on appearance alone. This is the same gap that exists at the examination gate: the receipt communicates a claim about payment with no mechanism by which that claim can be confirmed against its source. A receipt photocopied with an altered name and RRR number is indistinguishable from a genuine one to an examiner without programmatic verification access (Okoye & Ezeanyanso, 2013). The purple paper issued on the basis of that receipt therefore inherits the fraud risk of the document it was checked against. Synchronisation delays between Remita's systems and any university-side records compound the problem: a recent legitimate payment may not appear on a printed list, while a reversed transaction may not yet have been

flagged. The only mechanism capable of resolving these uncertainties definitively is a live API query against Remita's own records at the point of credential issuance.

2.2.3 QR Code Technology

QR codes are two-dimensional matrix barcodes originally developed by Denso Wave in 1994 for automotive parts tracking (Rouillard, 2008). They can encode up to 4,296 alphanumeric characters in a compact square format readable by any smartphone camera, without dedicated scanning hardware (Denso Wave, 2020; Kumar & Walia, 2011). Reed-Solomon error correction allows a code to remain scannable even with up to 30% of its surface obscured or damaged, which makes QR codes durable in physical environments like printed examination passes (Law & So, 2010).

In educational settings, QR codes have been used for resource linking, attendance tracking, and certificate authentication (Bonifacio, 2012). Their fundamental vulnerability in security contexts is that a photographed or screenshotted code is functionally identical to the original: any device can scan it. This means a code without cryptographic protection and server-side state tracking is reusable by anyone who photographs it. Addressing this requires encrypting the payload so the content cannot be read or fabricated without access to the server's private key, signing the ciphertext so tampering can be detected, and maintaining a database record of whether the token has been consumed (Chen et al., 2013).

2.3 Review of Related Works

2.3.1 Evolution of Verification Systems

Student verification has evolved from manual paper registers through computerised lookup systems to barcode-based ID cards, each transition improving lookup speed without fundamentally changing the attack surface (Tan et al., 2011; Sommerville, 2016). Barcode systems required dedicated scanning hardware, limiting adoption in resource-constrained environments. The smartphone revolution changed the calculus: QR codes offer higher data

capacity than barcodes and require no equipment beyond the cameras already carried by most staff, making them accessible in contexts where procurement of dedicated hardware is a barrier (Yusuf, 2020).

2.3.2 International QR Verification Systems

The Singapore Examinations and Assessment Board implemented QR codes on examination entry passes in 2015, enabling tablet-based examiner scanning that reduced verification times and improved record accuracy (Singapore SEAB, 2015). The system demonstrates QR viability for high-stakes examination admission but requires institutional investment in tablets for all venues and does not incorporate payment verification, which is irrelevant to Singapore's education funding model but central to the Nigerian context.

The University of Cambridge uses QR codes on examination certificates to link to a secure verification portal, allowing employers and institutions to confirm credential authenticity after the fact (Cambridge Assessment, 2020). This addresses post-examination credential validation, not admission control, and provides no model for real-time payment integration. Korean universities have deployed time-based QR codes for lecture attendance, with expiry enforced by timestamp rather than database state (Kim & Lee, 2018). Time-based expiry creates a window during which codes can be copied and reused; database-enforced one-time consumption closes that window entirely.

2.3.3 Nigerian Digital Verification Initiatives

The University of Lagos implemented a QR code event management system for seminar attendance that demonstrated functional viability within Nigerian infrastructure constraints (Nwankwo, 2018). The system operates in low-stakes contexts and carries no payment verification or cryptographic security, but it confirms that QR scanning works reliably in the Nigerian operational environment. Covenant University's Student Information Management System incorporates online registration, payment tracking, and digital ID cards with embedded QR codes (Bankole & Babalola, 2020). The QR codes in that system link to web profiles rather

than carrying cryptographically secured self-contained payloads, leaving them vulnerable to screenshot sharing. Connectivity failures have also periodically forced reversion to manual processes during examinations, which at Covenant University, as at most Nigerian institutions, means returning to the physical document-based clearance workflow.

The Remita-University Integration Project, implemented across several federal universities, demonstrates real-time API-based payment confirmation through programmatic queries against Remita's records (Ibrahim & Ibrahim, 2015). This infrastructure directly supports this project's payment verification layer. The gap is that no Nigerian institution has yet extended that API capability to the point of credential issuance, meaning the Remita query that could be made at registration is not made, and a printed receipt that cannot be verified replaces it at the faculty building and again at the examination gate.

2.3.4 Security-Focused Research

Chen et al. (2013) proposed an authentication protocol for QR code applications using encrypted payloads and digital signatures, demonstrating that forging valid codes without access to the server's cryptographic keys is computationally infeasible. Their work focuses on commercial applications and does not address payment integration, but it provides the security framework this project adapts to the examination context. Yeh et al. (2011) developed a two-factor QR code authentication system for repeated online logins, showing how QR codes fit into layered security architectures; however, their one-time-use handling was designed for session-based re-authentication rather than the strict single-consumption model required for examination admission. Sharma and Singh (2015) analysed vulnerabilities in online examination systems, identifying weak authentication, inadequate encryption, and poor audit trail maintenance as the primary failure modes, and recommending strong encryption, comprehensive activity logging, and multiple verification factors. Their security principles apply directly to this project's design even though their context is online test delivery rather than physical admission.

2.3.5 Comparison of Related Systems

Table 6 summarises the systems reviewed, their key characteristics, and their relevance to this project.

System	Key Features	Limitations	Relevance to This Project
Singapore SEAB	QR exam entry passes; tablet scanning	Requires tablets; no payment integration	Proves QR viability for high-stakes admission
Cambridge Certificate Verification	QR codes on certificates link to portal	Post-exam only; not real-time	Demonstrates QR authentication in education
Korean Attendance Systems	Time-based QR codes via smartphone	Time expiry weaker than DB enforcement	Proves smartphone scanning in classrooms
UNILAG Event QR System	QR codes for seminar attendance	Low-security; no payment verification	Confirms local technical feasibility
Covenant University SIMS	Integrated student services with digital ID	QR codes not cryptographically secured	Shows integration potential and challenges
Remita API Integration	Real-time payment verification via API	Not extended to exam admission control	Provides payment verification infrastructure
Chen et al. (2013)	Encrypted QR codes with digital signatures	Commercial focus; no payment integration	Provides cryptographic security framework
Sharma & Singh (2015)	Security analysis of exam systems	Online exam delivery focus	Informs security requirements and audit design

Table 6: Comparison of related systems, their limitations, and their relevance to this project.

2.4 Theoretical Frameworks

2.4.1 Role-Based Access Control

Role-Based Access Control (RBAC) assigns system permissions according to user roles rather than individual identities (Ferraiolo & Kuhn, 1992). In this system, students can register and

retrieve their own QR codes but cannot scan other students' codes or access administrative functions. Examiners can scan codes and view results but cannot modify student records or generate tokens. Administrators can manage accounts and view session reports but cannot override cryptographic validation. This separation ensures that compromising one account type does not expose the functions of another, and it makes the permission structure auditable without reference to individual user histories (Stallings & Brown, 2018).

2.4.2 One-Time Token Authentication

One-time tokens are authentication credentials that become permanently invalid after a single successful use, preventing replay attacks where a captured credential is reused by an unauthorised party (Hevner et al., 2004). Each QR code in this system contains a unique token identifier that is atomically marked as USED in the database at the moment of examiner approval. Any subsequent scan of the same code, whether by the original student or by someone who photographed the printed pass, encounters a USED status and is rejected. The enforcement is at the database level rather than within the QR code itself, which means the physical printability of the code does not create a reuse vulnerability. This is a stronger guarantee than time-based expiry, which creates a window of reuse risk between generation and expiry (Chen et al., 2013).

2.4.3 Cryptographic Security

Two cryptographic primitives underpin the QR token security model. AES-256-GCM symmetric encryption encodes the student payload so that intercepting or photographing a QR code yields only ciphertext; the data cannot be read without the server's private key. The GCM mode also provides authenticated encryption, meaning any modification to the ciphertext causes decryption to fail with an authentication error rather than silently producing corrupted output (Stallings & Brown, 2018). HMAC-SHA256 digital signatures provide origin authentication: each token carries a cryptographic fingerprint computed using the server's signing key that confirms the token was generated by the legitimate system and has not been tampered with since generation. Forging a valid signature without the signing key is computationally infeasible at current attack

capabilities. Together, these mechanisms ensure that QR codes can only be produced by the server, their contents cannot be altered, and their stated identity cannot be read without authorisation (Yeh et al., 2011).

2.4.4 Authentication and Authorization

Authentication and authorisation address different questions: authentication confirms identity (is this person who they claim to be?), while authorisation determines access rights (is this person permitted to enter?). Conflating them is a common design error in verification systems. In this project, QR code scanning and photograph matching handle authentication, confirming that the person presenting the code is the legitimate student whose data is encoded within it. Authorisation is handled separately through the payment record: a student whose identity is authenticated but whose payment has not been verified is not admitted. The two-stage model is more robust than systems that treat eligibility as a single undifferentiated question, because a failure in either stage produces a distinct, actionable outcome rather than a single ambiguous reject (Ferraiolo & Kuhn, 1992).

2.5 Gaps in Existing Literature and Systems

Four gaps in existing research and deployed systems justify the design of this project.

First, no existing system addresses the structural fragmentation created by a two-stage manual clearance process in which identity and payment are checked at different physical locations by different people using different documents. At Adekunle Ajasin University, the faculty building step and the examination gate step each represent incomplete verification: the faculty building checks the Remita receipt but issues a handwritten purple paper with no photograph and no cryptographic link to the payer; the examination gate checks the purple paper but has no mechanism to confirm it was legitimately issued for the person presenting it. No reviewed system has been designed to collapse both stages into a single digitally verifiable credential issued at registration, which is the specific contribution this project makes (Okoye & Ezeanyanso, 2013; Petters & Okon, 2014).

Second, most existing QR verification research originates in well-resourced contexts and does not evaluate systems against developing-country constraints. Systems requiring dedicated tablets, consistent high-speed connectivity, or centrally managed hardware procurement face adoption barriers in Nigerian university environments that the research does not account for (Yusuf, 2020). This project is explicitly designed around the infrastructure that already exists: staff smartphones, campus printers, and standard web hosting.

Third, time-based code expiry, the most common mechanism for preventing QR code reuse in reviewed systems, is weaker than database-enforced one-time consumption. A code that expires in ten minutes can be reused by anyone who photographs it within that window. A code that is atomically consumed upon first approval cannot be reused regardless of timing. This distinction is often conflated in the literature but represents a material security difference (Chen et al., 2013).

Fourth, existing systems do not address the operational lifecycle of examination verification infrastructure across multiple sessions and academic periods. The ability to invalidate all tokens from a completed session, clear usage state, and prepare the system for the next examination is a practical requirement for institutional deployment that is largely absent from published research (Nwankwo, 2018).

2.6 Conclusion

The literature establishes that the component technologies required for this project are individually mature and proven: QR codes have been successfully deployed in verification contexts across multiple continents, Remita provides programmatic access to authoritative payment records, and established cryptographic protocols for encrypted and signed tokens provide a well-tested security foundation. What has not been done is bringing these components together into a unified system designed to replace the specific two-step manual clearance process, faculty building visit followed by handwritten purple paper, that defines the current examination experience at Nigerian universities. The gaps identified in this review, particularly

the structural fragmentation between the faculty building and examination gate stages, the absence of database-enforced one-time use, and the lack of attention to Nigerian operational realities, define the contribution this project makes. The theoretical frameworks of RBAC, one-time token authentication, AES-256-GCM encryption, and the authentication-authorisation distinction inform every major design decision presented in Chapter Three.

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